Numpy Assignment Two

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1. What is the difference between population and a sample?

A **population** is the entire group of individuals, items, or data points that you want to study and draw conclusions about. It includes every member of the group that meets your specified criteria.

For example:

All students at Delhi University

Every smartphone user in India

All households in Mumbai city

A **sample** is a subset of the population that you actually collect data from and analyze. Since studying an entire population is often impractical or impossible, researchers use samples to represent the population.

For example:

500 randomly selected students from Delhi University

1,000 smartphone users surveyed across India

200 households randomly chosen from different Mumbai neighbourhoods

2. What is difference between inferential and descriptive statistics?

Descriptive Statistics summarize and organize data from a sample to describe its main features. They provide a clear picture of what the data shows without making broader conclusions.

Examples:

Mean, median, mode (measures of central tendency)

Standard deviation, variance, range (measures of spread)

Histograms, bar charts, box plots (visualizations)

Frequency distributions

Inferential Statistics use sample data to make predictions, estimates, or draw conclusions about the larger population. They help you generalize findings beyond your sample.

Examples:

Hypothesis testing (testing if a drug works)

Confidence intervals (estimating population parameters)

Regression analysis (predicting outcomes)

ANOVA (comparing group differences)

3. **A set of samples have been collected from a larger sample and the sample mean values are. 12.8, 10.9, 11.4, 14.2, 12.5, 13.6, 15, 9, 12.6. Find the population mean.**

```
import numpy as np

sample_mean = np.array([12.8, 10.9, 11.4, 14.2, 12.5, 13.6, 15, 9, 12.6])

add = sum(sample_mean)

n = len (sample_mean)

population_mean = add/n

print(population_mean)
```

4. **What is the significance of Exploratory Data Analysis (EDA)?**
    a. Data Quality Assessment: Helps identify missing values, errors, and inconsistencies in your dataset

    b. Pattern Discovery: Reveals hidden relationships, trends, and structures that might not be obvious initially

    c. Assumption Checking: Verifies if your data meets the assumptions required for advanced statistical methods

    d. Hypothesis Generation: Suggests potential hypotheses that can be formally tested later

    e. Decision Making: Guides decisions about which statistical methods to use and how to preprocess data

5. **What Are Skewness and Kurtosis?**

   Skewness measures the asymmetry of a probability distribution:

   a. Positive Skew (Right-skewed): Tail extends to the right, mean > median > mode.Common in income data, where few very high values pull the mean right.
   b. Negative Skew (Left-skewed): Tail extends to the left, mean < median < mode. Common in retirement age data, where few very young retirements pull the mean left
   c. Zero Skew: Perfectly symmetrical distribution (like normal distribution)

Kurtosis measures the tail ness of a distribution:

   a. Positive Kurtosis (Leptokurtic): Heavy tails, sharp peak - more extreme outliers than normal distribution
   b. Negative Kurtosis (Platykurtic): Light tails, flat peak - fewer extreme values
   c. Zero Kurtosis (Mesokurtic): Normal distribution pattern

6. **What is Outlier?**

   An outlier is a data point that significantly differs from other observations in the dataset. It's an extreme value that lies outside the overall pattern of the data.